

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project: DemoQA Book Store Application

Module: Login

Document ID: SRS-DEMOQA-LOGIN-001

Version: 1.0

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1. Introduction

1.1 Purpose

This document defines the basic functional and non-functional requirements for the **Login Module** of the DemoQA Book Store Application.

1.2 Scope

The Login module allows registered users to enter their credentials and access the application.

2. Functional Requirements

FR-01: Username Input

The system shall provide a username field where the user can enter their username.

Example:

Username: testuser

FR-02: Password Input

The system shall provide a password field where the user can enter their password.

The password should be displayed in masked form.

Example:

Entered: Password123

Displayed: ••••••••

FR-03: Login

The system shall provide a **Login** button to submit the entered credentials.

FR-04: Successful Login

When valid credentials are entered, the system shall authenticate the user and allow access to the authenticated area.

Example:

Valid Username + Valid Password → Successful Login

FR-05: Invalid Login

When invalid credentials are entered, the system shall reject the login attempt and display an appropriate error message.

Example:

Valid Username + Wrong Password → Login Rejected

FR-06: Empty Fields

The system shall handle empty username and/or password fields appropriately.

Examples:

- Empty username + password
- Username + empty password
- Both fields empty

FR-07: Logout

After successful login, the user shall be able to log out of the application.

After logout, the authenticated session should no longer provide access to protected content.

FR-08: Unauthorized Access

A user who is not authenticated shall not be able to access protected authenticated content directly.

3. Non-Functional Requirements

NFR-01: Usability

The Login page shall provide clearly identifiable:

- Username field
- Password field
- Login button
- Error/validation messages

NFR-02: Security

The system shall:

- Reject invalid credentials.
- Mask password input.
- Prevent unauthorized access.
- Invalidate the authenticated session after logout.

NFR-03: Compatibility

The Login module should function correctly on supported browsers such as:

- Chrome
- Firefox
- Edge

NFR-04: Performance

The Login operation should provide a reasonable response under normal network conditions.

Actual response time will be recorded during testing rather than assuming an unverified fixed limit.

4. Business Rules

ID	Business Rule
BR-01	Only users with valid credentials can authenticate successfully.
BR-02	Invalid credentials must not provide authenticated access.
BR-03	A logged-out user must not retain authenticated access.

5. Requirement Traceability

The requirements will be mapped to test scenarios and test cases.

Requirement → Test Scenario → Test Case → Result → Defect

Example:

FR-04 Successful Login
→ **TS-01 Valid Login**
→ **TC-01 Login with valid credentials**
→ Pass/Fail
→ BUG-XXX if failed

6. Assumptions and Constraints

- Valid test credentials will be available for testing.
 - The application will be accessible during testing.
 - Requirements not defined in this SRS will not be assumed as mandatory.
 - Detailed backend/database implementation is outside the scope of this SRS.
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7. Requirement Summary

ID	Requirement	Type
FR-01	Username input	Functional
FR-02	Password input/masking	Functional/Security
FR-03	Login button	Functional
FR-04	Successful login	Functional
FR-05	Invalid login handling	Functional
FR-06	Empty-field handling	Functional
FR-07	Logout/session termination	Functional/Security
FR-08	Unauthorized access prevention	Security
NFR-01	Usability	Non-functional
NFR-02	Security	Non-functional
NFR-03	Browser compatibility	Non-functional
NFR-04	Reasonable login performance	Non-functional
BR-01	Valid credentials required	Business Rule
BR-02	Invalid credentials rejected	Business Rule
BR-03	Logout removes authenticated access	Business Rule